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CMPUT 365: Introduction to Reinforcement Learning

Worksheet 12: Policy Gradient Methods

Question 1. Consider the episodic MDP that starts in state s_0. The actions a_1, a_2 are available in s_0: a_1 deterministically moves the agent to s_1 with reward 0, a_2 moves the agent to the terminal state, $\perp$, with reward 0. The actions a_1, a_2 are also available in s_1: a_1 takes the agent to $\perp$ with reward 0, a_2 takes the agent to $\perp$ with reward 2.

The policy is represented as a softmax in each state: for state s,

$$\pi(a = i \mid s, \boldsymbol{\theta}) = \frac{e^{\boldsymbol{\theta}_{s,i}}}{e^{\boldsymbol{\theta}_{s,1}} + e^{\boldsymbol{\theta}_{s,2}}},$$

with parameter vector $\boldsymbol{\theta} = (\theta_{0,1}, \theta_{0,2}, \theta_{1,1}, \theta_{1,2})$. Start at $\boldsymbol{\theta}^{(0)} = (0, 0, 0, 0)$. Use REINFORCE (episodic, $\gamma = 1$, $\alpha = 0.1$):

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha \sum_{t=0}^{T-1} G_t \nabla_{\boldsymbol{\theta}} \ln \pi(A_t \mid S_t, \boldsymbol{\theta}).$$

You observe the single episode

$$s_0 \xrightarrow{A_0=a_1,\ R_1=0} s_1 \xrightarrow{A_1=a_2,\ R_2=2} \perp$$

(a) Compute the gradients $\nabla_{\boldsymbol{\theta}} \ln \pi(A_t \mid S_t, \boldsymbol{\theta}^{(0)})$, returns G_0, G_1, and the parameter update after this episode (no baseline). Give $\boldsymbol{\theta}^{(1)}$.

(b) Repeat using a global baseline $b = 1$ (replace G_t by $G_t - b$). Give $\boldsymbol{\theta}^{(1)}$.

(c) Briefly compare the two updates and explain how the baseline affected magnitude/sign.

(a) At $\boldsymbol{\theta}^{(0)}$ every action probability is 0.5. Returns are $G_1 = 2$, $G_0 = 2$. Now let us compute the gradients.

$$\pi(a = i \mid s, \boldsymbol{\theta}) = \frac{e^{\boldsymbol{\theta}_{s,i}}}{e^{\boldsymbol{\theta}_{s,1}} + e^{\boldsymbol{\theta}_{s,2}}}, \qquad \ln \pi(a = i \mid s, \boldsymbol{\theta}) = \boldsymbol{\theta}_{s,i} - \ln\left(e^{\boldsymbol{\theta}_{s,1}} + e^{\boldsymbol{\theta}_{s,2}}\right)$$

For the first term of $\ln \pi$ we obtain the Kronecker delta $\mathbf{1}\{j = i\}$:

$$\frac{\partial}{\partial \boldsymbol{\theta}_{s,j}} \boldsymbol{\theta}_{s,i} = \begin{cases} 1, & \text{if } j = i, \\ 0, & \text{if } j \neq i. \end{cases}$$

For the second term, we use the chain rule:

$$\frac{\partial}{\partial \boldsymbol{\theta}_{s,j}} \ln\left(e^{\boldsymbol{\theta}_{s,1}} + e^{\boldsymbol{\theta}_{s,2}}\right) = \frac{1}{e^{\boldsymbol{\theta}_{s,1}} + e^{\boldsymbol{\theta}_{s,2}}} \cdot e^{\boldsymbol{\theta}_{s,j}} = \pi(a = j | s, \boldsymbol{\theta}).$$

Combining both we then have:

$$\frac{\partial}{\partial \boldsymbol{\theta}_{s,j}} \ln \pi(a = i \mid s, \boldsymbol{\theta}) = \mathbf{1}\{j = i\} - \pi(a = j | s, \boldsymbol{\theta})$$

Now, we can compute the numerical values for the gradients:

$$\nabla_{\boldsymbol{\theta}} \ln \pi(a_1 \mid s_0, \boldsymbol{\theta}) = \begin{bmatrix} 0.5 \\ -0.5 \\ 0 \\ 0 \end{bmatrix}, \qquad \nabla_{\boldsymbol{\theta}} \ln \pi(a_2 \mid s_1, \boldsymbol{\theta}) = \begin{bmatrix} 0 \\ 0 \\ -0.5 \\ 0.5 \end{bmatrix}.$$

With $\alpha = 0.1$ and no baseline, each step multiplies its gradient by $\alpha \cdot G_t = 0.1 \cdot 2 = 0.2$, giving us:

$$\begin{aligned}
\boldsymbol{\theta}^{(1)} &= \begin{bmatrix} 0.0 \\ 0.0 \\ 0.0 \\ 0.0 \end{bmatrix} + 0.1 \cdot \sum_{t=0}^{1} G_t \nabla_{\boldsymbol{\theta}} \ln \pi(A_{t+1} | S_t, \boldsymbol{\theta}) \\
&= \begin{bmatrix} 0.0 \\ 0.0 \\ 0.0 \\ 0.0 \end{bmatrix} + 0.1 \cdot G_0 \cdot \nabla_{\boldsymbol{\theta}} \ln \pi(a_1 \mid s_0, \boldsymbol{\theta}) + 0.1 \cdot G_1 \cdot \nabla_{\boldsymbol{\theta}} \ln \pi(a_2 \mid s_1, \boldsymbol{\theta}) \\
&= \begin{bmatrix} 0.0 \\ 0.0 \\ 0.0 \\ 0.0 \end{bmatrix} + 0.1 \cdot 2 \cdot \begin{bmatrix} 0.5 \\ -0.5 \\ 0.0 \\ 0.0 \end{bmatrix} + 0.1 \cdot 2 \cdot \begin{bmatrix} 0.0 \\ 0.0 \\ -0.5 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 0.1 \\ -0.1 \\ -0.1 \\ 0.1 \end{bmatrix}
\end{aligned}$$

(b) With a global baseline $b = 1$ each effective return is 1, and each step multiplies its gradient by $\alpha \cdot (G_t - 1) = 0.1 \cdot 1 = 0.1$. Thus, if we were to do the same computation as above, we would obtain

$$\boldsymbol{\theta}^{(1)} = [0.05, -0.05, -0.05, 0.05]^\top.$$

(c) The baseline reduces the magnitude of the updates (here by a factor of $1/2$) and, more generally, reduces variance. If $G < b$, it can even flip the sign of the updates. This matters because the baseline can make all actions appear more desirable (if it increases the gradients), with the best action seeing the largest gradient—and thus probability—increase. Conversely, it can make all actions less desirable (if it decreases the gradients), with the best action experiencing the smallest gradient—and thus probability—reduction. Recall that the probabilities are re-normalized after each update.

Question 2. Exercise 2.9 from Sutton & Barto (2018).

Dr. Sutton has asked that the solutions to the textbook exercises not be made electronically available to students. You are encouraged to attend office hours, where the teaching assistants can discuss the exercises with you and show you the answers. Alternatively, if you email Dr. Sutton your answers for an entire chapter, he will likely send you the electronic answer sheet for that chapter.

Question 3. Exercise 13.1 from Sutton & Barto (2018).

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Question 4. Exercise 13.3 from Sutton & Barto (2018).

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Question 5. Exercise 13.4 from Sutton & Barto (2018).

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